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cpp include "mbed.h"
// ===== // 1. // =====
// — — DigitalOut segA(p13); DigitalOut segB(p14); DigitalOut segC(p15); DigitalOut segD(p16);
DigitalOut segE(p17); DigitalOut segF(p18); DigitalOut segG(p19); DigitalOut segDP(p20); DigitalOut
digitLeft(p23); DigitalOut digitMid(p22); DigitalOut digitRight(p21);
// — () — // DigitalOut ewred(p25); DigitalOutewyellow(p26); DigitalOutewgreen(p27);
// DigitalOut nsred(p28); DigitalOutnsyellow(p29); DigitalOutnsgreen(p30);
// — — DigitalOut pedblue(p24); // : DigitalOutpedwhite(p12); // :
// — (S1) — DigitalIn row1(p5, PullUp); DigitalOut col1(p9);
// ===== // 2. // =====
enum TrafficState STEWGO, //STEWWARN, //STNSGO, //STNSWARN, //STPEDWALK//;
TrafficState currentstate = STEWGO; intcountdown = 0; boolpedrrequest = false;
// () define TIMEEWGO15defineTIMEEWWARN3defineTIMENSGO10defineTIMENSWARN3defineTIME
Timer systimer; floatlasttick = 0;
// const int digitPatterns[10][7] = 1,1,1,1,1,0, 0,1,1,0,0,0, 1,1,0,1,1,0,1, 1,1,1,1,0,0,1, 0,1,1,0,0,1,1,
1,0,1,1,0,1,1, 1,0,1,1,1,1,1, 1,1,1,0,0,0,0, 1,1,1,1,1,1,1, 1,1,1,1,0,1,1 ;
// ===== // 3. // =====
void setSegments(int idx) if (idx > 0 — idx < 9) segA=0; segB=0; segC=0; segD=0; segE=0; segF=0;
segG=0; segDP=0; return; segA=digitPatterns[idx][0]; segB=digitPatterns[idx][1]; segC=digitPatterns[idx][2];
segD=digitPatterns[idx][3]; segE=digitPatterns[idx][4]; segF=digitPatterns[idx][5]; segG=digitPatterns[idx][6];
segDP=0;
void clearDisplay() digitLeft=1; digitMid=1; digitRight=1; segA=0; segB=0; segC=0; segD=0; segE=0;
segF=0; segG=0; segDP=0;
void refreshdisplay()intnum = countdown; if(num < 0)num = 0;
if (num <= 10) setSegments(num / 10); digitLeft = 0; waitus(2000); clearDisplay(); setSegments(num
bool isbuttonpressed()col1 = 0; waitus(20); boolpressed = (row1 == 0); col1 = 1; returnpressed;
void updateights()ewred = 0; ewyellow = 0; ewgreen = 0; nsred = 0; nsyellow = 0; nsgreen = 0; pedblue = 0; pedwhite = 0;
switch (currentstate)caseSTEWGO : ewgreen = 1; nsred = 1; pedwhite = 1; break; caseSTEWWARN : ewyellow = 1;
void processtrafficlogic()if(countdown < 0)switch(currentstate)caseSTEWGO : currentstate = STEWWARN; co
// ===== // 4. // =====
int main() systimer.start(); lasttick = systimer.read();
currentstate = STEWGO; countdown = TIMEEWGO; pedrrequest = false;
updateights();
while(1) float now = systimer.read(); if(now-lasttick >= 1.0)lasttick = now; countdown —; processtrafficlogic
refreshdisplay();
if (isbuttonpressed())waitus(200000);
if (currentstate! = STPEDWALK)pedrrequest = true;
if ((currentstate == STEWGO||currentstate == STNSGO)countdown > 5)countdown = 5;/52

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